



Stability Analysis and Genotype $\times$ Environment Interaction
Studies|Application of Best Linear Unbiased Prediction (BLUP) in
Stability Analysis and Genotype $\times$ Environment Interaction
Studies

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Abstract:

Genotype $\times$ environment interaction (GEI) is a major challenge in plant breeding, as it complicates the identification of genotypes that are both high-performing and stable across diverse environments. Multi-environment trials (METs) generate complex datasets, often unbalanced and incomplete, where traditional ANOVA or regression-based stability metrics are limited. Best Linear Unbiased Prediction (BLUP), within a mixed model framework, provides a robust approach for estimating genotypic performance while accounting for random effects, environmental variation, and heterogeneous residuals. BLUP enables shrinkage-based predictions, improving accuracy over raw means or BLUE, particularly in unbalanced datasets. Furthermore, incorporation of pedigree or marker-derived relationship matrices allows exploitation of genetic correlations, enhancing prediction of untested genotypes and facilitating genomic selection. BLUP-based methods can also quantify stability through variance of predicted genotypic values across environments, factor-analytic decomposition of $G \times E$, and reliability-adjusted indices, providing actionable insights for selection. This manuscript reviews theoretical foundations, practical implementation, and computational approaches for BLUP in METs, including hierarchical and factorial designs. It also discusses integration with factor-analytic models for complex GEI, the use of combined pedigree and marker information, and strategies for handling missing data. BLUP-based stability analysis represents a flexible and powerful tool for modern plant breeding, enabling selection of widely adapted and specifically adapted genotypes while maximizing genetic gain under diverse environmental conditions.

Keywords: Mixed models, Plant breeding, Multi-environment trials

Mathematics Subject Classification (2010): 62P10, 62J12, 62H25,

92D10.

1 Introduction

Plant breeders aim to select genotypes that combine high yield potential with stability across diverse environments. Phenotypic performance of genotypes is determined by their genetic makeup (G), environmental conditions (E), and the interaction between genotype and environment (G×E). While classical analysis of variance (ANOVA) partitions total phenotypic variation into G, E, and GEI components, it does not provide precise predictions of genotypic performance, especially in unbalanced trials or when data from multiple environments and years are available (Smith et al., 2001). Stability analysis is often employed to quantify the consistency of genotype performance across environments. Classical methods include the regression-based Finlay-Wilkinson approach, AMMI (Additive Main Effects and Multiplicative Interaction) models, and Eberhart-Russell parameters (Lynch & Walsh, 1998). However, these approaches have limitations in handling unbalanced data, missing observations, and complex pedigree structures.

Mixed models and Best Linear Unbiased Prediction (BLUP) have emerged as powerful tools for addressing these limitations (Piepho & Möhring, 2006). BLUP provides estimates of genotypic effects by explicitly modeling random effects and variance-covariance structures, accommodating unbalanced datasets, repeated measurements, and heterogeneous variances. Furthermore, BLUP can incorporate pedigree or marker-based genetic relationships, improving prediction accuracy for selection in breeding programs. This manuscript reviews the theoretical foundations of BLUP in the context of GEI and stability analysis, discusses model formulation for hierarchical and factorial experimental designs, and presents practical applications and considerations for plant breeding programs.

2 Stability Experiments

Multi-environment trials are used to evaluate the performance of genotypes across diverse environmental conditions. Typical designs include randomized complete block designs (RCBD), incomplete block designs, lattice designs, and factorial experiments. Trials are often conducted over multiple years and locations to capture environmental variability. Phenotypic data collected from METs are subject to both spatial and temporal variation (Fig. 1). The statistical model for phenotypic data is generally expressed as:

$$y_{ijk} = \mu + g_i + e_j + (ge)_{ij} + \varepsilon_{ijk}$$

BLUP is derived from the general linear mixed model (LMM):

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \varepsilon$$

$$\mathbf{u} \sim N(0, \mathbf{G}), \varepsilon \sim N(0, \mathbf{R})$$

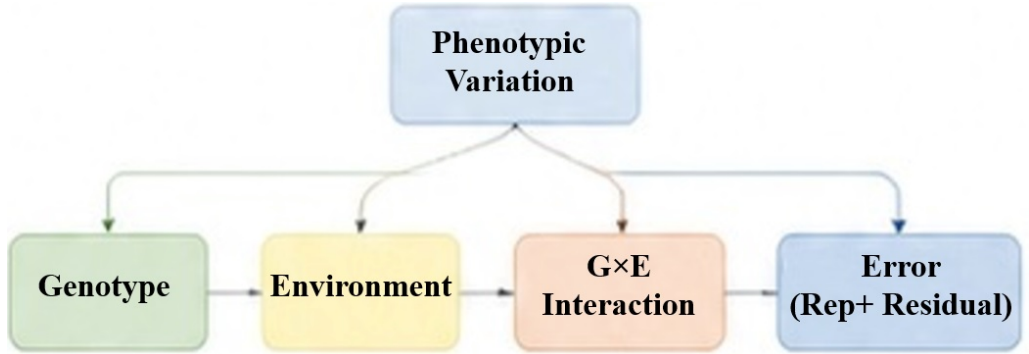
where $\mathbf{G}$ represents the variance–covariance structure among random effects, which can be simple (homogeneous) or complex (heterogeneous or structured by pedigree or markers). $\mathbf{R}$ represents residual covariance, which may also accommodate spatial or temporal correlation (Piepho & Möhring, 2006). The BLUP of $\mathbf{u}$ is obtained by solving the mixed model equations:

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{bmatrix}$$

Genetic relationships among genotypes can be included via the numerator relationship matrix ($\mathbf{A}$) or marker-based similarity matrices ($\mathbf{S}$). The variance-covariance structure of genotypic effects can be expressed as:

$$\mathbf{G} = \sigma_g^2 \mathbf{A} \text{ or } \mathbf{G} = \sigma_g^2 \mathbf{S}$$

$$\mathbf{G} = r_A^2 \mathbf{A} + r_S^2 \mathbf{S}$$



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Figure 1. Decomposition of phenotypic variance

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BLUP can be applied to hierarchical (nested) and factorial designs: Hierarchical (nested), so genotypes nested within families. Model allows estimation of within- and between-family effects. Factorial (crossed), like diallel crosses or hybrids from two heterotic pools. Model can include general combining ability (GCA), specific combining ability (SCA), and test-cross effects.

For example, a diallel model with GCA and SCA effects is expressed as:

$$y_{ijk} = \mu + g_i + g_j + s_{ij} + e_k + \varepsilon_{ijk}$$

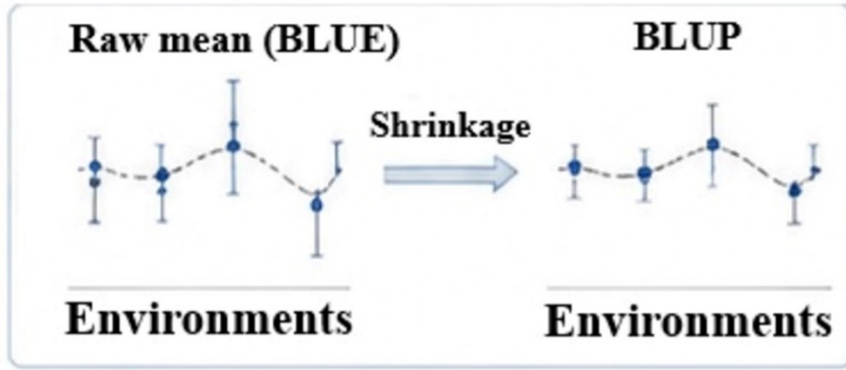
Stability is assessed by decomposing GEI effects or predicting genotype performance across environments with low fluctuations (Fig. 2).

BLUP-based approaches included (i) BLUP of genotype-by-environment effects which predicts genotypic performance in specific environments (Schabenberger & Gotway, 2005). (ii) Reliability-based stability measures so every stability variance can be estimated using predicted genotypic values. (iii) Factor-analytic models, which approximate G×E covariance structure using latent factors; BLUPs of factor scores provide stability estimates.

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Figure 2. Effect of shrinkage on raw means

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Simulation can illustrate BLUP performance. For example, an experiment with 50 genotypes tested in 6 environments, with 3 replicates per environment, can be simulated with heterogeneous genetic variances. BLUP provides shrinkage-adjusted predictions with higher correlation to true genotypic values than raw means or BLUEs, particularly in unbalanced datasets.

3 SAS codes

Regarding accept the assumptions of Model 1, the effect of the general adaptability ability (GAA) of genotype i is equal to $u+vi$. The deviation from the sum $u+vi$ represents the specific adaptability ability effect (SAA_{ik}) of genotype i in environment k . This effect consists of a linear component (the effect of environment dk) and a nonlinear component (the interaction effect $(vd)_{ik}$). To determine the effects of GAA_i and SAA_{ik} , the mean phenotypic values across replications are calculated. In this way, the results of testing n genotypes across m environments can be arranged in an $n \times m$ table, where: x_i is the sum of all phenotypic values for genotype i , x_k is the total phenotypic sum in environment k , $X_{..}$ is the grand total of all phenotypic values, x_{ik} is the phenotypic value of genotype i in environment k . The GAA and SAA effects are calculated using the following formulas (Fig. 1):

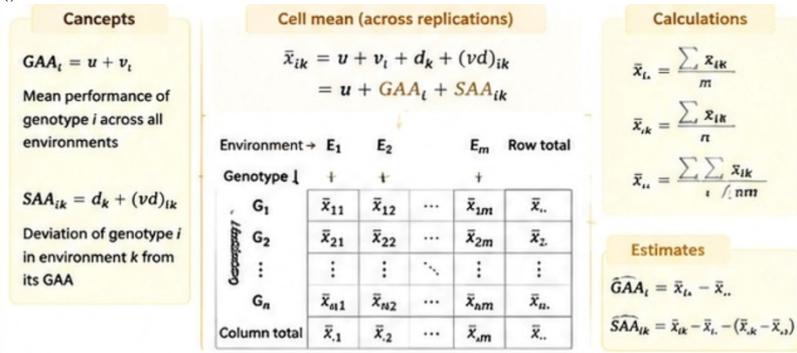
$$x_{i.} = \sum_k x_{ik} = x_{i1} + x_{i2} + \dots + x_{im}$$

$$x_{nk} = \sum_k x_{ik} = x_{1k} + x_{2k} + \dots + x_{nk}$$

$$x_{ik} = u + v_i + d_k + (vd)_{ik}$$

$$x_{ik} = u + GAA_i + SAA_{ik}$$

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Figure 1. Schematic illustration of stability evaluation

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The variance of random deviations for the mean phenotypic value in a specific environment x_{ik} is computed as:

$$var_{x_{ik}} = \frac{\sigma_e^2}{c} = \sigma^2$$

The variance of the difference between the means of two genotypes is determined as:

$$var(x_{ik}/x_{nm}) = 2\sigma^2$$

To obtain the standard error of the differences between effects, the square root of the corresponding variance must be taken. The least significant difference (LSD) is then obtained by multiplying the standard error by a constant magnitude. Comparison of genotypes in terms of general adaptability can be performed by comparing GAA_i values. To estimate the interaction strength of genotype i with environments, the genotype $\times$ environment interaction variance can be used. To measure the stability of genotype i , it is recommended to use the variance of SAA_i . Therefore, stability can be defined as the capacity of a genotype to maintain its phenotypic expression under various environmental conditions through organized mechanisms.

4 Stability analysis

Practical code examples for fitting BLUP in multi-environment trials

(METs), including stability analysis using factor-analytic models are:

```
proc mixed data=MET_data;  
class Genotype Environment Replicate;  
model Yield = Environment / solution;  
random Genotype Genotype*Environment;  
/* Genotype and GxE effects as random */  
lsmeans Genotype / pdiff cl;  
run;
```

Genotype and Genotype*Environment are treated as random effects to obtain BLUPs.

Environment is fixed to capture environmental means.

lsmeans Genotype provides predicted genotypic values across environments.

Practical code examples for fitting BLUP in multi-environment trials

(METs), including

Factor-analytic model for $G \times E$ are:

```
proc mixed data=MET_data method=REML;  
class Genotype Environment;  
model Yield = Environment;  
random Genotype / type=fa(1) g;  
repeated Environment / subject=Genotype type=ar(1);  
ods output SolutionR=BLUPs;  
run;
```

type=fa(1) specifies a factor-analytic covariance structure for $G \times E$.

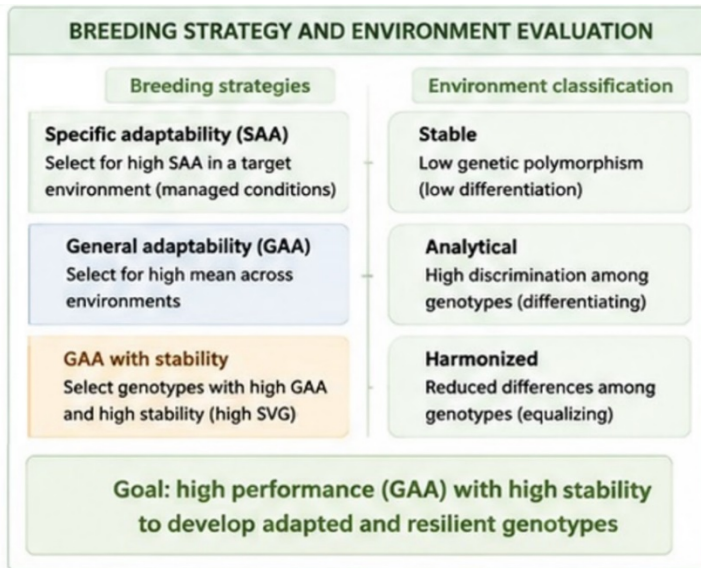
repeated allows modeling residual correlation across environments.

BLUPs output contains predicted genotypic values adjusted for $G \times E$.

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Figure 2. Flowchart of breeding and selection strategy

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stability]Breeding for GAA while considering stability

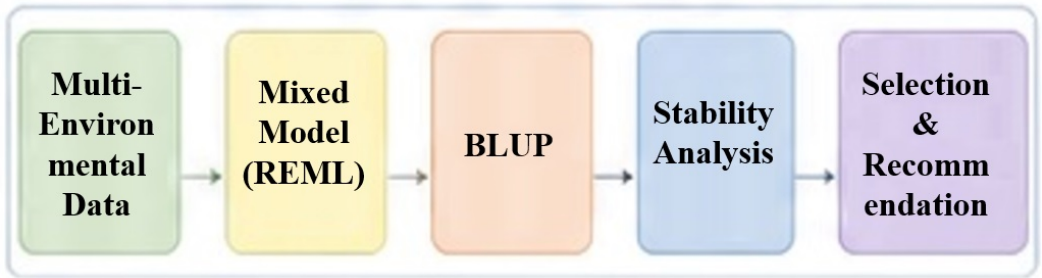
The BLUP accounts for missing data, unbalanced replication, and heterogeneous variance, yielding more accurate estimates of genotypic performance across environments compared to classical ANOVA or regression-based stability analysis. In factorial and diallel crosses, BLUP allows estimation of GCA and SCA effects even for untested hybrids (Bernardo, 1994). BLUP predictions enable the derivation of stability metrics across environments, so variance of predicted genotype performance across environments reflects stability. Also, factor-analytic models separate main effects from interaction patterns, highlighting genotypes with stable performance (low sensitivity to environmental variation) or specific adaptation (high performance in certain environments).

BLUP can model complex covariance structures among environments, such as autoregressive or spatially correlated residuals (Edwards & Jannink, 2006). Genotypic effects can be predicted for untested environments, enabling selection for wide or targeted adaptation. Marker information can enhance BLUP predictions by improving estimation of genetic relationships, especially for genotypes with incomplete pedigree data. Combined models of A and S matrices allow optimal utilization of both pedigree and marker data.

BLUP provides a flexible, robust framework for analyzing MET data and

assessing GEI. Compared with classical stability metrics, BLUP improves prediction of genotypic performance under unbalanced and incomplete data (Fig. 3). Also, it accounts for complex variance-covariance structures, including spatial, temporal, and pedigree effects. BLUP Supports hierarchical and factorial designs, enabling dissection of GCA, SCA, and within-family variation. Finally, it Facilitates integration of pedigree and marker-based genetic relationships for genomic prediction. The main assumptions of BLUP are normally distributed random effects and correctly specified variance-covariance structures. In practice, REML estimation is robust to moderate deviations from normality, although outliers may affect inference (Piepho & Möhring, 2006). Accurate variance component estimation is crucial, particularly in plant breeding, where the number of genotypes is often limited. Bayesian or shrinkage approaches can help address heterogeneity among crosses or environments (Edwards & Jannink, 2006). BLUP-based stability analysis also allows flexible interpretation of GEI (Schabenberger & Gotway, 2005). Factor-analytic models and prediction-based stability metrics provide information on general adaptability and specific adaptation, guiding breeders in selecting genotypes for wide or targeted environmental adaptation.

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Figure 3. Workflow for BLUP-based stability analysis

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5 Conclusion

BLUP is a powerful tool for stability analysis and GEI studies in plant breeding. By incorporating random effects, variance–covariance structures, and pedigree or marker information, BLUP enables precise prediction of genotypic performance across environments. Its ability to handle unbalanced and incomplete data, coupled with flexible model specification, makes it superior to classical methods in

multi-environment trials. The integration of BLUP with genomic information and hierarchical experimental designs further enhances selection efficiency. Future research should focus on robust variance estimation in small datasets, robustness to non-normality, and practical applications in modern breeding programs.

6 References

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